Engineering for Lean Manufacturing: A cross-functional process

Recently my firm has had the opportunity to consult for manufacturing engineering groups that engineer stamping plants, body shops, and final assembly lines within a major auto maker. This firm, like its competitors outside Japan, has been working to implement its version of the Toyota Production System, aka lean manufacturing. They started in the plants with pilot projects on existing operations. More recently they have been moving upstream to manufacturing engineering. In this company, like other large companies, manufacturing engineering starts at the division level for all new programs which are eventually handed off to small manufacturing engineering groups within the plants.

Lean manufacturing starts with the ideal of continuous flow. The goal is to eliminate as much as possible non-value added activities that prevent a one-piece flow of product. In an existing operation, a lean transformation group is faced with many existing conditions counter to lean. These can include so-called "monuments"—large batch processing machines that produce too much to meet the immediate demand for specific products. Other problems are equipment organized by process rather than product, layouts that impede smooth flow, equipment that has not been mistake proofed, and on and on. In a new operation there is an opportunity to develop the process lean from the start.

What does it take to do lean engineering? First, there are many different considerations essential to engineer a system for lean manufacturing. Second, generally these considerations are handled by separate cross-functional groups and unless these groups all work as a team it will be impossible to develop a truly lean system.

As an example, in a final assembly line a central issue is getting parts to the assembler on the line with minimal handling in the position that allows the assembler to attach the part to the vehicle with minimum wasted motion. This means thinking about allocating space on the floor for small parts supermarkets, locating call parts depots near the point of use on the line, designing returnable packaging for small lots of parts that positions parts in a convenient orientation, well designed parts presentation racks that allow easy return of containers, and designing parts presentation/material handling equipment with ergonomics in mind. In the large corporation we are working with it is clear that, at the minimum, these tasks involve manufacturing engineers, plant engineers, material handling engineers, material logistics professionals, and ergonomists at the division level; manufacturing engineering, material logistics, and production managers at the plant level; and outside vendors. Traditionally division personnel handle only broad planning decisions and leave a lot of the detail to the plants. But in lean manufacturing the broad planning and detailed decisions are interrelated. For example, it is impossible to know how big a holding area for parts (i.e., supermarket) should be without knowing how much of each type of material will be stored there. The physical size of containers is also needed to size supermarkets and these should be the right size for getting all the parts needed line side with good ergonomics. Thus, the early task of laying out the plant depends on downstream decisions on packaging, part quantities, racks, and parts presentation lineside.

Manufacturing engineers involved in early stages must stay with the project longer instead of handing off to engineers in the plants.

Here are some of the many considerations in engineering lean manufacturing plants that require cross-functional teamwork:

1. Factory layouts that support continuous flow-- Keeping material moving requires that you organize machines around product lines. Ideally you locate them next to each other, in sequence, so material can flow through the processes one piece at a time.

2. Right-sized and reusable machines--Product-oriented layouts generally require smaller, dedicated machines as opposed to large, multipurpose machines. Dedicated does not mean inflexible. Remember that equipment can be designed to be reusable, with minimum modifications, for future products.

3. Quick changeover technologies--In an ideal lean manufacturing environment, you can make any combination of products needed to satisfy the customer on the same line with negligible changeover times. We've all heard about single-minute exchange of dies in stamping. You can apply similar principles to reduce the time needed for many common manufacturing-process changeovers.

4. Error-proofing devices--Errors create waste, which interrupts continuous flow. All machines should automatically detect abnormal conditions and either alert operators or shut down.

5. Visual controls--Whenever operators take part in the control cycle, use visual controls (e.g., lights to communicate equipment status, clear markings for preventative maintenance, easily interpreted gauges).

6. Maintainability--Equipment breakdowns impedes continuous flow. Preventative maintenance programs are absolutely essential and should be facilitated by design for maintainability.

7. Ergonomics—Design tools, equipment, packaging, conveyor heights and racks to minimize stress on the operator whenever an operation requires manual intervention.

8. Engineering for teams—Most lean operations are built on teams of operators with considerable autonomy to control and continuously improve their operations. There are some physical engineering considerations that can facilitate that including designing for line-of-sight so team members and leaders can see each other and abnormal conditions, designing in team meeting facilities, placing related types of operations near to each other, designing u-shaped cells to facilitate communication and flexible staffing, keeping noise levels down, and locating comfort facilities near to the operations.

9. Person-machine separation—Simple automation can be used so that operators can move to the next operation while the current operation is cycling, such as automatic nut runners that snap on and tighten the nut to the proper torque automatically.

In the auto company we are working with engineers are having an enormous impact on lean manufacturing with impressive results. As one example, in one body shop engineers had laid out the facility with batch thinking in mind. This layout was reviewed by the newly formed lean engineering group. They noted that the door lines were separated from the assembly line where the doors were to be installed and were built in a batch mode and then moved to the line in large batches with fork lifts. The new team asked: How can we make this flow? With a little ingenuity they came up with a way to eliminate changeover time and build doors in the exact sequence needed on the assembly line with doors built and moved to the line on a conveyor one-by-one as they were needed. This new design resulted in dramatic reductions in inventory, a reduction of 8 operators, savings of almost 30,000 square feet of floor space, and a return on investment of about 35% in less than two years.

Engineering of new plants and production lines change dramatically when companies go lean. To do it right require a cross-functional effort. The size of the team depends on how large and complex the organization is to begin with. Obviously smaller companies have fewer specialized functions than large auto manufacturers and coordination will be easier, but still necessary. Engineering need to work closely with production, vendors, quality, and even finance. Cost justification of lean systems must consider a broader range of benefits than direct labor savings that are often the focus of ROI calculations. Engineering has just become a little more complicated, but potentially a lot more rewarding.

by:Jeffrey K. Liker
Author: *Becoming Lean: Experiences of US Manufacturers*
Productivity Press, 1997.
Director, Value Chain Analysis Program
Industrial and Operations Engineering
University of Michigan
Ann Arbor, MI
And Senior Lean Consultant, Optiprise, Inc.